E. Applications on the non-conservation of the quantity of movement

1. Braking of a vehicle

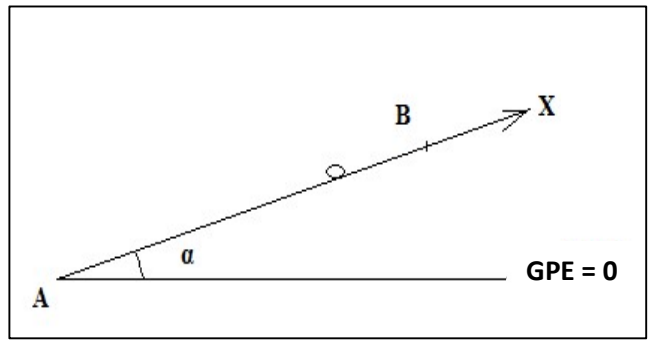
A car of mass M moves on a rough horizontal road at a velocity $\vec{V}_0$. Suddenly the driver acts on the brakes. The car stops at time t_1. Frictional forces are represented by a constant force $\vec{f}$, parallel to the road and opposite to the movement.

a. Express t_1 in term of f, f', M and V_0.

b. Express the traveled distance D in term of M, V_0, f and f'.

2. Translation on an inclined plane

A ball of mass m = 500g is thrown on an inclined plane of α = 0.1 rad relative to the horizontal with a speed $V_0 = V_A$ = 3m/s. The ball reaches point B with a speed V_B = 1m/s during Δt = 1.2s.



a. Applying the general form of Newton's second law, determine the intensity of the frictional force.

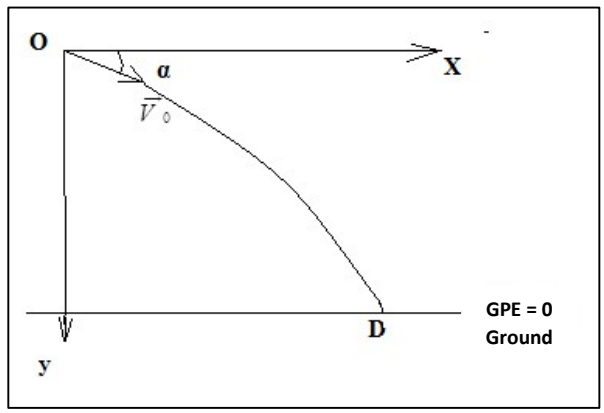
b. Determine the distance D traveled by the ball.

c. Calculate the amount of heat produced. Convert the answer found to calories.

3. Projectile shot with an oblique velocity $\vec{V}_0$

A projectile of mass m = 100 g is thrown from a point O, with a speed V_0 = 10 m/s at 30° with the horizontal, as shown in the figure. Friction is supposed to be negligible.

Given: h = 1m



a. Determine the speed of the projectile at point D just before reaching the ground.

b. Determine the P_x and P_y components of the momentum vector at time t.

c. Deduce the expressions of V_x and V_y at the instant t.

d. Determine the time equations of motion x (t) and y (t).

e. At what moment does the projectile reach the ground in D?

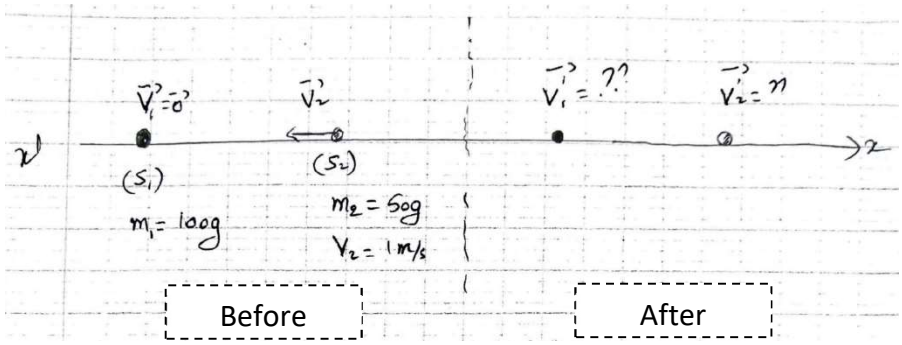
F. Applications on the conservation of linear momentum.

Application1: Elastic collision

a) Determine the velocities $\vec{V'}_1$ and $\vec{V'}_2$ of the two particles just after the collision.

b) If the collision takes places during Δt = 0.05 s, determine $\vec{F}_{1/2}$, the force exerted by (S1) on (S2).

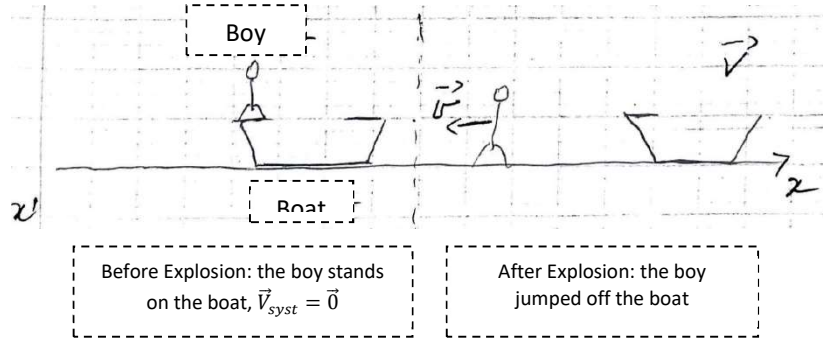
c) Deduce $\vec{F}_{2/1}$.



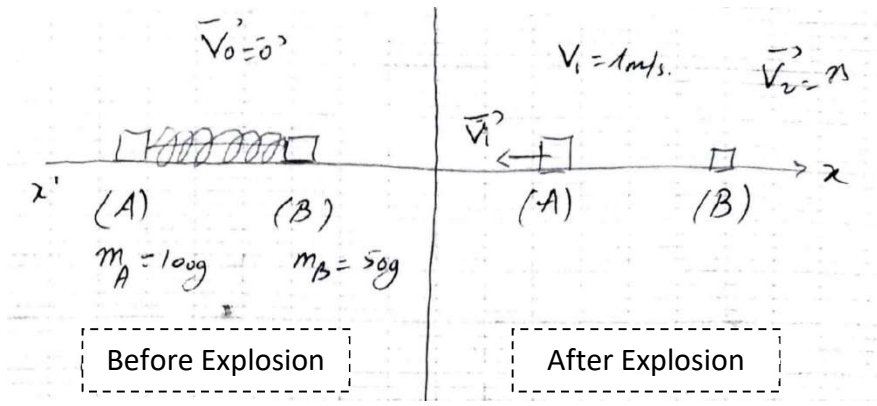
Application2: Jumping off a boat (explosion)

a) Determine $\vec{V}$, the boat's velocity in term of m, M, and $\vec{v}$.

b) Is the explosion elastic? Justify.



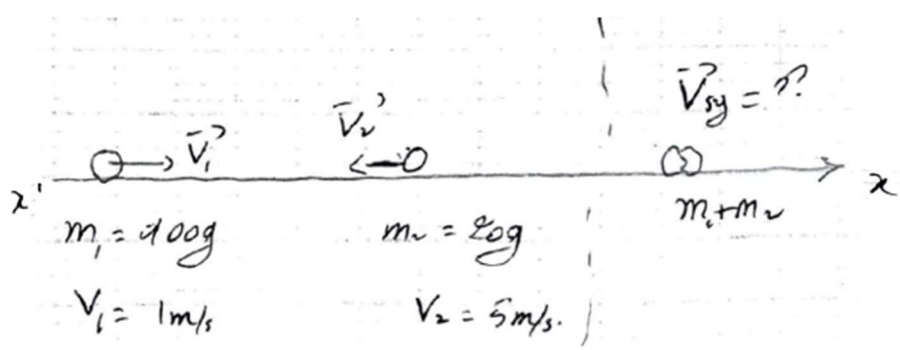
Application3: Explosion of two pucks



Two pucks (A) and (B) placed on an air table, are connected together by a wire. A spring (R) of negligible mass is placed between the pucks. As result (R) becomes compressed from L_0 = 10 cm to L = 5 cm. We burn the wire; the puck then flies away at $\vec{V}_1$ and $\vec{V}_2$.

a) Determine $\vec{V}_2$.
b) Is the explosion elastic? Justify.
c) Determine the spring constant K.

Application4: Colliding objects stick together



a) Determine $\vec{V}_{syst}$.
b) Does the KE increase or decrease during collision? Justify.